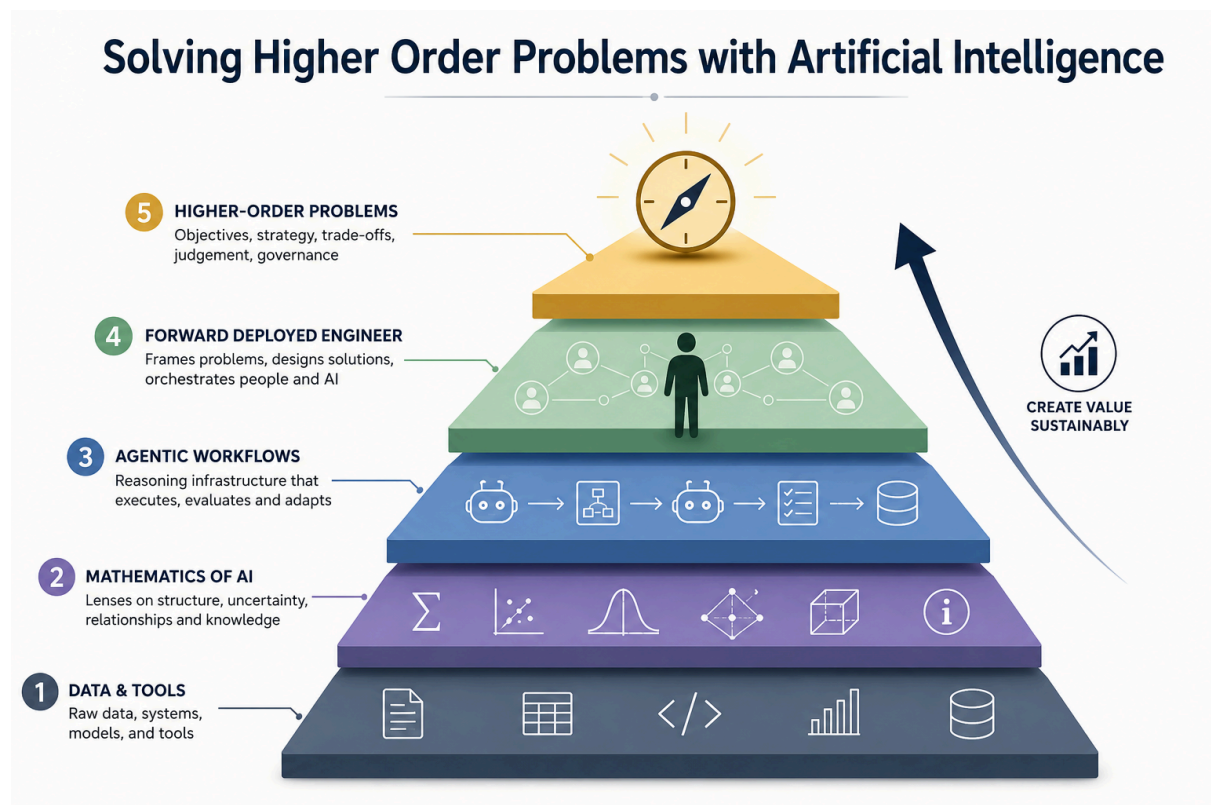


ERDOS RESEARCH · POSITION PAPER

Solving Higher-Order Problems with Artificial Intelligence

Euclid, Forward Deployed Engineers, and Agentic Workflows



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Abstract

Artificial intelligence is collapsing the cost of execution. Code, analysis, translation, summarisation and research — the tasks that once absorbed most professional effort — are increasingly performed by machines at negligible marginal cost. This does not dissolve the need for expertise. It relocates it. Value migrates upward, towards the problems that determine what should be executed at all: framing, objective-setting, trade-off management, coordination, governance and adaptation. We call these higher-order problems.

This paper argues that higher-order problems have a structure, and that the structure is tractable.

It advances four propositions.

First, that Euclid's method — definitions, postulates, common notions, propositions and proofs — remains the most reliable discipline available for reasoning under conditions of ambiguity, and maps directly onto the challenges of enterprise AI.

Second, that the Forward Deployed Engineer is the organisational role in which that discipline is exercised: not a builder of models but an architect of organisational intelligence.

Third, that agentic workflows, correctly designed, do not solve higher-order problems but constitute the reasoning infrastructure through which humans solve them at scale.

The fourth proposition is the one most often missed. The mathematics that underpins artificial intelligence — the questions of why learning is possible at all, what a representation captures, what mechanisms drive change, what principles govern a system, what remains unknown, and what intervention creates value — is not merely the theory behind the models. It is a general method for reasoning about complex systems. Linear algebra, probability, statistics, geometry and information theory function, at the higher-order level, as lenses on structure, uncertainty, evidence, relationships and knowledge. AI is best understood as the case study through which that method is learned, not as its subject.

Taken together, these four elements — a method, a role, an infrastructure and a set of analytical lenses — form an operating model for the AI-native enterprise. The paper concludes with failure modes and ten recommendations for organisations building this capability.

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Part I · The Four Elements Defined

Any serious treatment of a problem should begin by saying what the problem is. This paper concerns four things that are frequently invoked and rarely defined: higher-order problems, the Forward Deployed Engineer, agentic workflows, and the mathematics of artificial intelligence understood as a method rather than a body of theory.

Part I defines each, and then explains why they belong in the same argument.

In future, we intend to extend this idea to some other elements for the teaching of AI

1.1 Higher-order problems

A useful working definition is this:

Higher-order problems are problems concerned with selecting objectives, managing trade-offs, coordinating systems and exercising judgement under uncertainty. They sit above task execution and determine what should be done, why it should be done, and how success should be measured.

The distinction is easiest to see through the questions each type of problem asks. A first-order problem asks: how do we do the task? A higher-order problem asks: what task should we do, why, under what constraints, with what trade-offs, and how will we know it is working?

An equivalent characterisation is diagnostic rather than definitional. A problem is higher-order when the following conditions obtain.

SIX CONDITIONS

The problem itself is unclear — the statement of it is part of what is in dispute.

There are multiple stakeholders, whose objectives do not coincide.

There are feedback loops, so that the intervention changes the system it is measuring.

The environment changes over time, so a correct answer decays.

There is no single correct answer, only defensible positions on a trade-off surface.

Success depends on judgement, not solely on optimisation.

Problems meeting this description are not exotic. They are the problems that matter most: national AI strategy, climate change, enterprise transformation, healthcare reform, scientific discovery, building an AI-native company, designing AI governance, developing advanced AI safely. In none of these does the difficulty lie in execution.

A simple hierarchy

It is helpful to think in three levels.

Level	Problem type	Representative examples	Current status
Level 1	Execution problems — performing a specific, specifiable task	Write code Generate a report Translate a document Classify an image	Increasingly automated
Level 2	Workflow problems — coordinating multiple tasks into a reliable sequence	Design a support process Build a delivery pipeline Orchestrate several agents	Increasingly orchestrated
Level 3	Higher-order problems — deciding what the workflow should be and why	Which customers to serve first Which system to build What risk is acceptable How responsibility is allocated	Requires human judgement

The boundary between these levels is not fixed, and that is precisely the point. Every year, work migrates downward: what required judgement in 2023 is a workflow in 2025 and an execution task in 2026. The strategic question for any organisation is not whether this migration will occur but where its people will stand when it does.

The seven characteristics

Higher-order problems typically exhibit one or more of seven characteristics. The list is useful diagnostically: if a problem presented as technical exhibits three or more of these, it is not a technical problem.

Characteristic	The question it raises
Problem framing	What problem are we actually solving?
Goal definition	What does success mean, and how would we recognise it?
Trade-offs	What must be balanced against what, and who decides?
Uncertainty	What do we not know, and what follows if we are wrong?
Coordination	Who must act together, and in what sequence?
Governance	Who is accountable when the system errs?
Adaptation	How should the system change as conditions change?

A mathematical perspective

The distinction has a compact formal expression. A conventional optimisation problem asks us to find the value of x that maximises a known objective function $f(x)$. A higher-order problem asks a different question entirely: what should $f(x)$ be?

Given $f(x)$, find x → First-order → Given the world, choose $f(x)$ → Higher-order

This is not an academic distinction. A substantial proportion of AI failures in enterprise settings are not failures of optimisation but failures of objective selection: the system performed exactly as specified, and the specification was wrong. A recommendation engine optimised for engagement that degrades trust, a triage model optimised for throughput that displaces clinical risk onto the discharge pathway, a fraud model optimised for precision that quietly excludes a customer segment — each is a technical success and an institutional failure.

THE GOVERNING ASYMMETRY

First-order errors are visible, local and correctable. Higher-order errors are invisible, systemic and compounding. An organisation that optimises the wrong objective with great technical skill will fail faster than one that optimises the right objective badly.

The same distinction across domains

Domain	First-order problem	Higher-order problem
Software engineering	Write the function	What system should be built, and what trade-offs between speed, cost, security and reliability are acceptable?
Healthcare	Diagnose the patient	How should a system allocate constrained capacity, and which patients receive priority?
AI governance	Evaluate the model	What constitutes acceptable performance, which risks matter most, and who is accountable?
Business strategy	Build the product	Should this market be entered at all, and against which future scenarios is the model robust?
Public policy	Draft the guidance	What outcome is the regime intended to secure, and what behaviour will it actually produce?

1.2 The Forward Deployed Engineer

The Forward Deployed Engineer is conventionally described as a technical specialist embedded with a customer to help them implement a product. That description was adequate when the difficulty lay in implementation. It is no longer adequate, because implementation is the part that is being automated.

The role that matters in an AI-native environment is different in kind:

A Forward Deployed Engineer is a specialist who converts higher-order organisational problems into executable systems by translating domain expertise, designing workflows, orchestrating AI capabilities, governing risk and creating continuous learning loops.

Three features of this definition deserve emphasis. The FDE works at the boundary between an organisation's tacit expertise and its executable capability — this is the substance of the Expertise-to-Execution Stack. The FDE is accountable for the system, not the component: a model that performs well inside a workflow that fails is not a success. And the FDE operates on

live organisations rather than on specifications, which means the work is as much diagnostic and political as it is technical.

THE FDE IS DEFINED BY POSITION, NOT BY TOOLING

The FDE stands where the customer's problem is still ambiguous and the organisation's capability is still tacit. Everything downstream of that position — model selection, prompt design, evaluation harnesses, deployment — is engineering. Everything at that position is judgement. The role exists because the judgement cannot be shipped as a product.

1.3 Agentic workflows

An agentic workflow is a system in which one or more AI agents pursue a stated goal by selecting and sequencing their own actions, invoking tools, evaluating intermediate results and escalating to humans under defined conditions. The distinguishing feature is not autonomy but delegation of control flow: in a conventional pipeline, the sequence of steps is fixed by the designer; in an agentic workflow, the sequence is determined at run time in response to what is discovered.

It is useful to locate agents within the wider capability stack, since much confusion arises from treating an agent as a starting point rather than as the top of a ladder.

Layer	What it provides	What it does not provide
Prompts	A single reliable interaction with a model	Repeatability across people and cases
Skills	Encapsulated, reusable expertise with a defined contract	Integration with systems of record
Hooks	Connection to organisational data, tools and events	Control over sequence and failure handling
Harnesses	Evaluation, guardrails, observability and constraint	Goal selection
Agents	Run-time selection of actions in pursuit of a goal	Judgement about which goals are worth pursuing

The final row is the crux of this paper. Each layer of the stack extends what can be delegated, and no layer delegates the question of what should be delegated. That question is higher-order by construction, and it returns to a human every time.

1.4 The mathematics of artificial intelligence

The mathematics of AI is conventionally treated as the theory required to understand the models: the linear algebra behind an embedding, the probability behind a loss function, the information theory behind a compression bound. That treatment is correct and incomplete. The same mathematics, asked at a higher level of abstraction, constitutes a general method for reasoning about complex systems — one that applies to a national strategy or a health system exactly as it applies to a transformer.

The method can be stated as six questions. Each has a narrow form, in which it concerns machine learning, and a general form, in which it concerns any complex system.

#	The AI question	The higher-order question
1	Why can AI learn at all?	Is there hidden structure in this problem?
2	What does AI learn?	What representation best captures the reality?
3	How does AI learn?	What mechanisms drive change in this system?
4	Why does the mathematics work?	What principles govern the problem?
5	What do we not understand?	Which of our assumptions might be wrong?
6	How do we use AI?	What interventions actually change the system?

Part V develops this in full. For the purposes of definition, the point is that the mathematics supplies something the other three elements do not: a set of analytical lenses that determine what one is able to see. Euclid governs the form of the reasoning; the mathematics governs its content.

1.5 Why the four belong together

These elements are usually discussed separately — Euclid in the history of mathematics, the FDE in commercial delivery, agentic workflows in technical architecture, the mathematics of AI in a graduate curriculum. Treating them together is the argument of this paper, because each supplies what the others lack.

Element	Contribution	Failure without it
The Euclidean method	A discipline for moving from confusion to structured clarity	Confident conclusions resting on undefined terms and hidden assumptions
The mathematics of AI	Lenses that reveal structure, uncertainty, evidence, relationships and information	Rigorous reasoning conducted on the wrong representation of the problem
The Forward Deployed Engineer	A role that carries method and lens into a live organisation	Method that exists in documents and never touches the operating reality
Agentic workflows	Infrastructure that executes, explores and remembers at scale	Judgement that is sound but slow, unscaled and dependent on individuals



Part II · The Euclidean Method Applied to Higher-Order Problems

[Euclid](#) was not primarily teaching geometry. He was demonstrating a method for reasoning reliably about complex objects — a method whose ambition was that every conclusion in a body of knowledge should be traceable to something explicitly stated at the outset. That ambition is precisely what higher-order problems require and precisely what most enterprise reasoning lacks.

The Elements opens with definitions, then postulates, then common notions, and only then propositions and their proofs. Stated in modern terms:

Before solving a problem, establish what exists, what is assumed, what principles are accepted, and how conclusions will be validated.

Most higher-order problems fail at the first three steps. Organisations proceed directly to propositions — build the platform, deploy the agent, publish the policy — and discover eighteen months later that the participants were never talking about the same thing. What follows are ten principles drawn from the Elements, each stated as a Euclidean principle and applied to contemporary practice.

2.1 Definitions — clarify what you are talking about

Euclid begins by defining his objects: a point is that which has no part. The content of that particular definition matters less than the discipline it embodies — define the concept before reasoning about it.

Consider an organisation that states an intention to build responsible AI. Euclid's first question would be: what is responsible? Fairness? Transparency? Safety? Regulatory compliance? Explainability? These are distinct properties with distinct measurements and frequently conflicting implications. Without definition, every participant believes agreement has been reached while each is reasoning about a different object.

EUCLIDEAN PRINCIPLE

Ambiguous concepts produce ambiguous conclusions. The cost of the ambiguity is not paid at the point of definition; it is paid at the point of delivery.

2.2 Postulates — make assumptions explicit

Euclid states his assumptions openly and does not prove them: a straight line may be drawn between any two points. They are accepted, but they are accepted visibly.

An AI strategy typically rests on assumptions of comparable weight: that adoption will increase, that data quality can be improved within the programme's horizon, that staff will accept the workflow change, that the regulatory position will remain stable. These are rarely stated. They are load-bearing nonetheless, and when one fails, the programme fails without anyone being able to say why.

EUCLIDEAN PRINCIPLE

Every higher-order problem rests on assumptions. Good reasoning does not eliminate them — it makes them visible, so that their failure is detectable.

2.3 Common notions — establish shared principles

Euclid's common notions are the universal principles on which the whole edifice rests: things equal to the same thing are equal to each other. Their organisational analogues are the axioms of governance — accountability should be traceable, decisions should be evidence-based, risk controls should be proportionate to exposure.

Stating these explicitly resolves an unexpectedly large share of strategic disagreement. Many debates that present as disputes about tactics are in fact disputes about first principles, and they cannot be settled at the tactical level because the disagreement is not there.

EUCLIDEAN PRINCIPLE

Shared principles reduce disagreement. Where disagreement persists, it identifies the axiom in dispute.

2.4 Incremental construction — build from validated foundations

The Elements is remarkable because every theorem depends only on definitions, postulates and previously proved theorems. Nothing appears from nowhere.

The characteristic enterprise AI failure is the inverse. An organisation moves from AI strategy to AI platform to AI agents without establishing the data foundations, evaluation practice, governance and workflow understanding on which each depends. The resulting structure is impressive in presentation and unsupported in fact; it collapses at the first contact with an edge case, and the collapse is attributed to the technology.

Data foundations → Evaluation → Workflow → Governance → Agents

EUCLIDEAN PRINCIPLE

Complex systems should be built from validated foundations. A capability that cannot name what it rests on does not rest on anything.

2.5 Proof before scale

Euclid does not accept a proposition because it appears obvious. He proves it. The assertion that agents will improve productivity is a proposition of exactly this kind: plausible, widely believed, and unproven in the specific setting where it is to be relied upon. The Euclidean response is not scepticism but evidence — experiments, measurements, evaluation results, on the organisation's own work.

This is the methodological core of the Anti-Pilot position: the purpose of an early deployment is not to demonstrate that the technology functions but to establish, under conditions resembling production, whether the proposition holds.

EUCLIDEAN PRINCIPLE

Evidence precedes confidence. Scale amplifies whichever of the two you actually possess.

2.6 Diagrams help, but they do not prove

One of the deepest lessons of the Elements is its treatment of figures. Diagrams accompany the propositions and aid comprehension; the proof, not the diagram, establishes the result. A figure that looks right can be wrong, and the history of geometry contains celebrated cases in which it was.

The modern equivalents are architecture diagrams, demonstrations and strategy slides. Each is persuasive; none is evidence. A demonstration establishes that a system can succeed on a case chosen by the demonstrator, which is a much weaker claim than the one the audience typically takes away.

EUCLIDEAN PRINCIPLE

Visual intuition is useful and insufficient. Ask what would have to be true for the picture to be wrong, and then measure it.

2.7 Decomposition — break complexity into tractable components

Euclid resolves difficult constructions by reducing them to smaller established statements. The question how do we govern AI is not answerable as posed. Decomposed, it becomes a set of tractable questions: data governance, model governance, agent governance, evaluation, monitoring, accountability. Each admits of an owner, a control and a test.

EUCLIDEAN PRINCIPLE

Break complexity into components small enough to be owned, tested and refuted.

2.8 Traceability — every conclusion has a lineage

In the Elements, any theorem can be traced backwards through the propositions it depends on to the definitions and postulates at the origin. Nothing is mysterious, and nothing is unaccountable.

Decision → Reasoning → Evidence → Data → Policy

This is precisely the structure that modern AI assurance seeks to impose, and it is the structure that regulators increasingly require. A decision artefact that records the reasoning, the evidence supporting it and the policy under which it was taken is a Euclidean object: it makes the lineage of a conclusion inspectable after the fact.

EUCLIDEAN PRINCIPLE

Every conclusion should have a lineage. A decision whose lineage cannot be reconstructed cannot be governed, defended or improved.

2.9 Internal consistency

The Elements forms a coherent system; a contradiction anywhere indicates an error somewhere. Organisations are rarely audited to this standard. A company that declares a commitment to innovation while operating an approval regime that forbids experimentation has not made a trade-off — it has stated a contradiction, and the contradiction will be resolved informally, unpredictably and in favour of whichever pressure is more immediate.

EUCLIDEAN PRINCIPLE

Check for contradictions. A significant share of strategic failure is inconsistency that was never surfaced as such.

2.10 The meta-level insight

The deepest lesson of the Elements is not any particular proposition. It is this:

Do not ask only whether a conclusion is correct. Ask whether the system that produced the conclusion is sound.

This is the definitive test of a higher-order problem. A conclusion that happens to be right, produced by a process that could as easily have produced a wrong one, is not a solution. It is a coincidence with a favourable outcome — and it will not survive the next case.

2.11 The mapping

Euclid	Higher-order problem solving
Definitions	Clarify concepts before reasoning about them
Postulates	Make assumptions explicit and testable
Common notions	Establish shared principles and organisational axioms
Propositions	Form hypotheses that can be stated and refuted
Proofs	Validate with evidence gathered under realistic conditions
Incremental construction	Build capability on validated foundations
Decomposition	Reduce complexity to ownable components
Traceability	Maintain the lineage of every consequential decision
Consistency	Detect and resolve contradictions
Axiomatic thinking	Design systems whose soundness can be examined

2.12 A Euclidean protocol for higher-order problems

The method can be operationalised as an eight-step sequence. It is deliberately ordered: each step is answerable only once the preceding one has been settled, and the common pathology is to begin at step four.

Step	Question	Artefact produced
1 • Definitions	What exactly are we trying to achieve, and what do the key terms mean?	Definition set with agreed measurements
2 • Assumptions	What are we assuming that we have not proved?	Assumption register with failure indicators
3 • Principles	What values and constraints govern the decision?	Stated principles and non-negotiables
4 • Hypotheses	What candidate solutions might work, and why?	Hypothesis set with predicted effects
5 • Evidence	How will we test them, on what data, at what cost?	Evaluation design
6 • Proof	What result would justify commitment at scale?	Evidence pack and decision threshold
7 • Traceability	Can we reconstruct how we reached the conclusion?	Decision artefact
8 • Learning	What changes if an assumption proves false?	Revision triggers and review cadence

If Aristotle supplies a framework for reasoning and systems thinking supplies a framework for complexity, Euclid supplies a framework for structured certainty. Higher-order problems can never be made wholly certain. What Euclid offers is the passage from confusion to clarity: define carefully, assume explicitly, reason systematically, validate rigorously, build incrementally. That is as pertinent to AI governance, agentic architecture and enterprise strategy today as it was to geometry twenty-three centuries ago.

One limitation should be stated plainly, because it determines the structure of the rest of this paper. The Euclidean method governs the form of reasoning: whether terms are defined, whether assumptions are visible, whether conclusions carry their lineage. It does not tell you which terms are the right ones to define, or which of the countless possible representations of a situation will make its structure visible. That is a question of content rather than form, and it is where the mathematics of Part V does its work.

Part III · How Forward Deployed Engineers Solve Higher-Order Problems

Method without a role is literature. This part describes the eight functions through which a Forward Deployed Engineer converts higher-order problems into systems that operate. Each function is characterised by a transformation: something the organisation possesses in an unusable form becomes something it can act on.

3.1 Problem framing

Many organisations do not have a technology problem. They have a problem-definition problem. The client states a requirement for an AI chatbot; the higher-order question is what business problem the chatbot is intended to solve. The answer is frequently long customer wait times, poor knowledge management or inconsistent support quality — three different problems with three different solutions, only one of which resembles a chatbot.

The FDE's first act is therefore reframing: moving from the stated solution back to the actual constraint. This is where the largest single source of programme value is created or destroyed, and it happens in the first fortnight.

Symptoms → Root causes

Technology requests → Business problems

3.2 Goal engineering

Most organisations cannot state what success would look like. Improve productivity, use AI, automate workflows — these are directions of travel, not objectives. They cannot be evaluated, which means they cannot be falsified, which means the programme cannot fail and therefore cannot learn.

The FDE converts direction into objective by specifying the measurement. Improve customer service becomes a defined set: response time, resolution rate, escalation frequency, satisfaction, cost per interaction — together with the thresholds that would constitute success and the trade rate between them.

NOTE ON MEASUREMENT

Specifying metrics is itself a higher-order act, because the metric becomes the objective the system optimises. The Euclidean discipline applies with full force: define the term, state the assumption that the metric proxies the outcome, and record the conditions under which the proxy would break.

3.3 Trade-off navigation

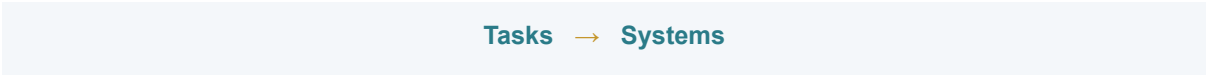
Real systems involve competing objectives that cannot be simultaneously maximised. The FDE's contribution is not to resolve these tensions by preference but to surface them as explicit, owned decisions.

Objective	In tension with	The decision that must be made explicit
Accuracy	Speed	What latency is acceptable, and for which classes of case?
Automation	Human oversight	Which decisions require a human in the loop, and on what trigger?
Cost	Reliability	What failure rate is tolerable at what unit economics?
Innovation	Governance	What may be tried without approval, and within what envelope?
Personalisation	Privacy	What data may inform a decision about an individual?

The transformation here is from competing priorities to explicit decisions. An unresolved trade-off does not disappear; it is resolved by default, invisibly, by whoever writes the implementation.

3.4 System design

Higher-order problems rarely sit in isolation. They involve people, processes, data, software and governance simultaneously. A hospital AI assistant is not a model producing an answer. It is clinicians, patients, electronic health records, governance, workflows, audit and escalation mechanisms — and the model is one component, frequently not the one that determines whether the system succeeds.



3.5 Knowledge externalisation

Most organisations depend on tacit knowledge held in individuals. That knowledge is unmeasurable, unscalable and departs when its holder does. The FDE's most durable contribution is to externalise it — to convert expertise into an organisational asset that persists independently of the expert.



This progression is the substance of the Expertise-to-Execution Stack. Its significance is that each step is inspectable: a skill can be reviewed, versioned, tested and governed in a way that a person's judgement cannot. The externalisation is never complete — the residue that resists capture is exactly the higher-order judgement this paper concerns — but the portion that can be captured compounds.

3.6 Human–AI coordination

Most organisations do not require full automation; they require an effective division of labour. The FDE specifies what humans do, what agents do, when escalation occurs and how decisions are validated. In claims processing, for example, the agent extracts information and assesses risk while the human reviews exceptions and takes the final decision — an allocation that is a design choice, not a technical constraint, and one that should be revisited as evidence accumulates.

Automation → Augmentation

3.7 Governance and risk design

A large share of AI failures are governance failures wearing technical clothing. The questions that determine outcomes are: who is accountable; what the agent is permitted to do; what data it may access; what evidence is required before it acts; and how its decisions are audited afterwards.

The critical design principle is that governance is built into the workflow rather than applied to it. A control that exists in a policy document and not in the execution path is not a control; it is an aspiration with a reference number.

Uncontrolled AI → Governed AI systems

3.8 Organisational learning

Most AI programmes fail because organisations do not learn systematically. The FDE installs two loops operating on different timescales. The operational loop — observe, evaluate, improve — governs the behaviour of a deployed system. The organisational loop — projects, lessons, patterns, standards — governs the behaviour of the institution that deploys them.

Observe → Evaluate → Improve

Projects → Lessons → Patterns → Standards

The second loop is the one organisations neglect, and it is the one that determines whether the tenth deployment is cheaper than the first. Without it, an organisation with fifty completed AI projects has fifty anecdotes rather than a capability.

3.9 The FDE higher-order problem framework

Higher-order problem	FDE contribution	Transformation
Problem framing	Define the real problem	Symptoms → root causes
Goal engineering	Define what success means	Vague goals → measurable objectives
Trade-offs	Balance competing objectives	Competing priorities → explicit decisions

Higher-order problem	FDE contribution	Transformation
System design	Design end-to-end workflows	Tasks → systems
Knowledge capture	Externalise expertise	Tacit knowledge → organisational capability
Human–AI coordination	Allocate responsibilities	Automation → augmentation
Governance	Manage risk and accountability	Uncontrolled AI → governed AI systems
Organisational learning	Create feedback loops	Projects → institutional memory

Historically, human value derived principally from execution. It increasingly derives from framing, judgement, coordination, adaptation and governance. This shifts the centre of gravity of the role upwards: the FDE becomes less a builder of components and more a capability architect — in effect, an architect of organisational intelligence.

AI does not eliminate expertise. It changes where expertise is applied — and the FDE is the role defined by that new location.

Part IV · How Agentic Workflows Solve Higher-Order Problems

The central claim of this part is deliberately restrictive:

Agentic workflows do not solve higher-order problems. They construct the conditions under which humans solve them — by exploring options, gathering evidence, coordinating activity, maintaining memory and learning continuously.

A compact statement of the division of labour: agents execute, workflows coordinate, humans judge. The power lies in the combination, and the characteristic error is to collapse the three into one and expect the agent to supply the judgement.

4.1 Why conventional workflow design fails here

Most workflows assume a determinate structure — input, process, output — which is adequate for well-defined problems. Higher-order problems have ambiguous goals, multiple stakeholders, genuine trade-offs, irreducible uncertainty and dynamic environments. They cannot be resolved by a single pass through a fixed pipeline, because the pipeline encodes an answer to the question that is actually at issue.

Input → Process → Output

The agentic alternative is not a longer pipeline but a different object — a decision-making system:

Goal → Multiple agents → Evaluation → Human judgement → Adaptation → Learning

4.2 Six patterns

Pattern 1 · Problem-framing agents

The most consequential failure in higher-order work is solving the wrong problem. When an executive asks how we should deploy AI, a framing workflow does not answer the question; it interrogates it. What business objective exists? Where is the bottleneck? What constraints apply? What alternatives were not considered?

Typical agents: discovery, stakeholder analysis, constraint mapping, opportunity identification. The output is not a recommendation but a structured problem statement — which is, in Euclidean terms, the definitions and postulates written down before anyone proceeds to propositions.

Pattern 2 · Multi-perspective reasoning

Higher-order problems require conflicting viewpoints to be held simultaneously. In a healthcare deployment, agents can be instantiated to reason from the position of the clinician, the patient, the administrator, the compliance officer, the engineer and the health economist. Each analyses the proposal independently; the workflow then synthesises areas of agreement, areas of conflict, risks and trade-offs.

This amounts to institutionalising debate — making adversarial review a routine, low-cost operation rather than an occasional and politically expensive one. Its value is not that the agents are as insightful as the professionals they represent, but that the conflicts surface early, when they are cheap to address.

Pattern 3 · Trade-off exploration

Where a decision involves competing objectives — accuracy against cost against speed against governance — an agentic workflow can generate multiple coherent scenarios and evaluate each against cost, risk, performance and strategic alignment. The workflow becomes a decision laboratory: the trade-off surface is explored systematically rather than argued about anecdotally, and the human chooses a point on a mapped surface rather than between two options that happened to be advocated.

Pattern 4 · Continuous evaluation

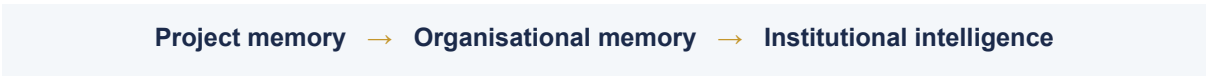
Higher-order problems rarely stay solved, because the conditions that made the answer correct continue to change. Agentic workflows can monitor performance, drift, emergent risk and business outcomes continuously. For a deployed support agent this means tracking hallucination rate, escalation rate, satisfaction and resolution time, and then acting on the trend.



This is the harness layer of the capability stack performing its proper function: not preventing the system from acting, but ensuring that the evidence needed to govern it is produced as a by-product of its operation.

Pattern 5 · Organisational memory

Most organisations solve the same problem repeatedly and learn from it once. Agentic workflows can capture decisions, the rationale behind them, the trade-offs accepted, the outcomes observed and the lessons drawn — and, critically, can retrieve them at the moment a similar decision arises.



This is among the most valuable and least implemented applications of agents. It is also the one that most directly serves the Euclidean requirement of traceability: an institution whose decisions carry their lineage can examine the soundness of the system that produced them, not merely the conclusions themselves.

Pattern 6 · Human–AI decision loops

Because higher-order problems require judgement, the workflow must be constructed so that judgement is exercised at the right point rather than absorbed by default into the automation.

Stage	Actor	Function
Generate	Agents	Produce candidate options with supporting evidence

Stage	Actor	Function
Choose	Humans	Select among options; own the trade-off and the accountability
Execute	Agents	Carry out the chosen course within defined constraints
Review	Humans	Assess outcomes against the objectives that were stated
Learn	Agents	Update memory, evaluation and workflow from the feedback

4.3 Worked example: enterprise AI strategy

The conventional approach engages consultants, runs workshops and produces a report — an artefact that is accurate on the day it is delivered and decays from that moment. The agentic alternative constructs a standing capability.

Component	Function
Discovery agent	Maps current AI usage, shadow adoption and existing capability
Process mining agent	Identifies where time, cost and error actually concentrate
Risk agent	Assesses governance exposure, regulatory duties and control gaps
Economics agent	Models cost, benefit and sensitivity under stated assumptions
Technology agent	Assesses architecture, data readiness and integration constraints
Change agent	Evaluates organisational readiness and adoption barriers
Synthesis agent	Reconciles the findings and produces options with trade-offs made explicit
Human steering group	Takes the decisions, owns them, and states the assumptions being accepted

The essential difference is not speed or cost. It is that the recommendations update as conditions change, and the reasoning behind them remains inspectable. The report becomes a system.

4.4 A general model

Any higher-order problem can be represented as a cycle of eight stages. The stages are not novel individually; what agentic infrastructure changes is the cost of executing them, and therefore the frequency with which an organisation can afford to.

Stage	Activity	Principal actor
Observe	Gather information from systems, people and the environment	Agents
Interpret	Construct explanations and identify what is actually happening	Agents with human review
Generate	Create candidate options	Agents

Stage	Activity	Principal actor
Evaluate	Assess options against objectives, risks and trade-offs	Agents and humans
Decide	Select a course and accept the accompanying risk	Humans
Execute	Carry out the decision within defined constraints	Agentic workflows
Learn	Compare outcome to expectation and record the difference	Agents and humans
Adapt	Revise goals, constraints and workflows accordingly	Humans

WHERE JUDGEMENT STAYS

Note the distribution. Agents dominate observation, generation and execution. Humans retain decision and adaptation — the two stages that determine what the system is for. An architecture that quietly reassigns those two stages to the agent has not automated a workflow; it has abdicated a responsibility that remains legally and institutionally yours.

4.5 The deeper insight: reasoning infrastructure

The most valuable use of agentic workflows is not the automation of tasks. It is the construction of what might be called a reasoning infrastructure for the organisation.

The analogy is instructive. ERP systems became infrastructure for transactions; CRM systems became infrastructure for customer relationships. Neither was primarily valuable because it performed clerical work faster. Each was valuable because it made a class of organisational activity visible, consistent and improvable. Agentic workflows can play the same role for judgement, coordination, learning, decision-making and adaptation.

The combination of Forward Deployed Engineers, agentic workflows and organisational memory moves AI from being a tool for execution to being a system for continuously addressing higher-order problems.

Part V · Using the Mathematics of Artificial Intelligence to Solve Higher-Order Problems

The preceding parts established a method, a role and an infrastructure. This part supplies the analytical content that runs through all three — and it does so from an unexpected source. The mathematics that makes artificial intelligence possible turns out, when its questions are asked at the right level of abstraction, to be a general methodology for higher-order problems.

The six-part framework is not really a framework for AI. It is a framework for solving higher-order problems under uncertainty. AI is the case study through which the method is learned.

The claim deserves scrutiny rather than acceptance, so this part works through each of the six questions in its narrow and general forms, then examines the five mathematical lenses that support them, applies the whole to a worked example, and closes with why this matters specifically to the Forward Deployed Engineer.

5.1 Question one — why can we learn about this problem?

FINDING STRUCTURE IN COMPLEXITY

The narrow question is why machine learning is possible at all. There is no mathematical guarantee that a function fitted to observed data will perform on data not yet seen; generalisation works because the world contains exploitable regularity, and because a model's inductive bias happens to align with it. Learning is possible when structure exists and the learner is built to find it.

The general question is the same one: is there hidden structure in this problem? The first act in any higher-order problem is to search for recurring patterns, binding constraints, incentives, dependencies and system dynamics. In healthcare reform, patients, clinicians, insurers and governments interact in ways that are neither random nor arbitrary; they respond to incentives with considerable regularity. The mindset is to assume that structure exists and to go looking for it — which is precisely the assumption underlying generalisation.

WHY THIS QUESTION COMES FIRST

If a problem genuinely has no exploitable structure, no amount of analysis will produce a reliable intervention, and the correct response is resilience rather than optimisation. Establishing that structure exists — and where — is what makes everything downstream worth doing. This is the same judgement a practitioner makes when deciding whether a problem is learnable before selecting an architecture.

5.2 Question two — what are we learning?

REPRESENTATION

The narrow question concerns what a model actually learns: not the training examples but a representation of them, in which distance and direction carry meaning. The reason embeddings transformed the field is that they changed the representation, not the objective.

The general question — what representation best captures this reality? — is arguably the most under-appreciated question in enterprise practice. Most difficult problems are representation problems in disguise. Climate change can be represented as emissions data, as an economic system, as an energy network, as a structure of political incentives, or as ecosystem dynamics. Each representation is defensible; each makes certain truths visible and others invisible; each implies a different set of available interventions.

THE CONTROLLING INSIGHT

Better representations create better solutions. A problem that appears intractable under one representation is often routine under another, and the effort spent finding the second representation returns more than the effort spent optimising within the first.

This connects directly to Euclid. Definitions are the act of choosing a representation, and doing it explicitly. When an organisation defines responsible AI as compliance rather than as safety, it has not merely clarified a term — it has selected a representation of the problem that determines which interventions will subsequently appear sensible.

5.3 Question three — how does change happen?

MECHANISMS

The narrow question concerns learning dynamics: how gradient descent moves a system through a loss landscape, why some architectures converge and others do not. The general question asks what mechanisms drive the system under study — its actors, decisions, incentives, feedback loops and modes of adaptation.

Why do cities grow? Why do some ventures succeed? Why do scientific breakthroughs occur when they do? The answers lie in mechanisms rather than in correlations, and the distinction is practical: a correlation tells you what to expect if nothing changes, whereas a mechanism tells you what will happen when you intervene. Higher-order problems are, by definition, problems in which someone intends to intervene.

5.4 Question four — what principles govern the system?

FIRST PRINCIPLES AND THE FIVE LENSES

The narrow question asks why the mathematics works. The general question asks what fundamental principles govern the problem — and it is here that the five mathematical pillars become directly usable as instruments of thought.

Mathematical lens	Higher-order lens	The question it asks
Linear algebra	Structure	What are the important dimensions, and which are we ignoring?
Probability	Uncertainty	What do we not know, and how is that uncertainty distributed?
Statistics	Evidence	What evidence should we trust, and how much should it move us?

Mathematical lens	Higher-order lens	The question it asks
Geometry	Relationships	How are the components connected, and what is near to what?
Information theory	Knowledge	What information would actually change a decision?

Applied as a sequence, these five questions constitute a rapid diagnostic that can be run against any difficult problem. Their power lies in exposing what has been omitted. A programme that has considered structure, relationships and evidence but never asked which information would change a decision is a programme that will collect data indefinitely without ever reaching a decision point — an outcome familiar to anyone who has observed an enterprise data strategy.

ON INFORMATION AS THE GOVERNING LENS

Information theory earns particular attention in the agentic era. The value of an additional evaluation, an additional monitoring signal or an additional review stage is not its cost but the extent to which it would change what is done. Instrumentation that cannot alter a decision is expenditure, not evidence — and agentic systems make it very easy to generate a great deal of it.

5.5 Question five — what do we not yet understand?

FRONTIER THINKING

The narrow question is the honest one that research keeps in view: what remains unexplained about why these systems work. The general question is its organisational equivalent — which of our assumptions might be wrong? What are we missing? Which variables are hidden? Which feedback loops are invisible? What observation would genuinely surprise us?

This is the scientific disposition, and it is where breakthroughs originate. It is also the question that organisational incentives most reliably suppress, because articulating it requires a senior person to state publicly that something material is not known. The Euclidean assumption register is the mechanism by which this question is made routine rather than heroic: if every load-bearing assumption is written down with its failure indicator, asking what might be wrong becomes an inspection rather than a confession.

5.6 Question six — what actions create value?

DECISION AND INTERVENTION

The narrow question asks how AI is used. The general question asks which interventions actually change the system. At this stage policy becomes action, strategy becomes execution and science becomes engineering, and the questions turn concrete: what should we do, what should we build, what should we optimise, and what should we measure?

This is the point at which the four elements of this paper converge. The mathematics has identified the structure and the leverage; the Euclidean protocol has established the evidence threshold; the Forward Deployed Engineer designs the intervention into the operating reality; and the agentic workflow executes, evaluates and remembers it. Value is created here and

nowhere else — but the quality of what is created here is determined entirely by the five questions that preceded it.

5.7 Worked example: AI governance through the six questions

Question	Applied to AI governance
1 · Structure	Why is governance needed here? Look for the recurring patterns in AI failure — misspecified objectives, distribution shift, unclear accountability — rather than treating each incident as unique
2 · Representation	What best captures governance: a set of policies, a capability model, a risk taxonomy, or the agentic workflows themselves? The choice determines what can be controlled
3 · Mechanisms	How do the governance mechanisms actually operate — monitoring, evaluation, human oversight, feedback loops — and which of them have teeth?
4 · Principles	Which mathematics applies: risk distributions, decision theory under uncertainty, information flow, network effects between interacting agents
5 · Unknowns	What remains genuinely unknown — emergent behaviour, agent-to-agent interaction, long-horizon risk — and what would we observe first if it were occurring?
6 · Interventions	What creates value: controls in the execution path, evaluations against real cases, telemetry that changes decisions, and a governance architecture that scales

Note what the sequence prevents. An organisation that begins at question six — the position most begin from — buys controls, evaluations and telemetry without having established which failures recur, which representation makes them visible, or which signals would change a decision. It will have a governance function that is expensive, busy and unable to say what it has prevented.

5.8 Why this matters to the Forward Deployed Engineer

A Forward Deployed Engineer is rarely solving a narrow optimisation problem. The problems that arrive are representation problems, uncertainty problems, coordination problems and decision problems — and the six-part framework provides a systematic route through them.

Observe → Represent → Understand → Explain → Challenge → Intervene

Stage	The question being answered	Where it appears in this paper
Observe	Is there structure here, and where?	Part III · problem framing
Represent	What representation makes the structure visible?	Part II · definitions
Understand	What mechanisms drive the system?	Part III · system design

Stage	The question being answered	Where it appears in this paper
Explain	What principles govern it, under which lenses?	Part V · the five lenses
Challenge	Which assumptions might be wrong?	Part II · assumption register
Intervene	What action creates value, and how is it governed?	Part IV · agentic execution

This is why mathematical foundations are properly understood as a leadership capability rather than a technical prerequisite. The purpose of studying them is not to enable an engineer to derive a gradient — that is now delegable — but to equip a practitioner to recognise, in an unfamiliar and politically charged situation, which dimensions matter, how uncertainty is distributed, what evidence should move them, how the parts are connected, and which piece of information would change what everyone does next.

A course in the mathematical foundations of AI is not merely teaching AI. It is teaching a way of thinking about complex systems, uncertainty, knowledge and intervention — with AI as the case study through which the method is learned.

Part VI · Synthesis, Failure Modes and Recommendations

6.1 The combined operating model

The four elements of this paper occupy distinct positions and are not substitutes. Euclid supplies the method — the discipline by which ambiguity is converted into structured, inspectable reasoning. The mathematics of AI supplies the lenses — the analytical instruments that determine which features of a situation become visible in the first place. The Forward Deployed Engineer supplies the role — the person who carries method and lens into a live organisation with its politics, its legacy systems and its tacit knowledge. Agentic workflows supply the infrastructure — the means by which the resulting reasoning is executed, evidenced, remembered and revised at a scale no individual could sustain.

Layer	Element	Question it answers	Failure if absent
Method	Euclidean reasoning	How do we reason soundly about this?	Confident conclusions on undefined terms
Lens	Mathematics of AI	What are we able to see, and what are we missing?	Rigour applied to the wrong representation
Role	Forward Deployed Engineer	Who converts the reasoning into a working system?	Method that never touches operations
Infrastructure	Agentic workflows	How does this operate, scale and persist?	Judgement that cannot outlive its author

6.2 A worked integration

Consider an organisation that states a requirement: deploy AI agents in customer operations. Applied in sequence, the four elements produce a materially different programme from the one that would otherwise result.

Euclidean step	FDE action	Agentic support
Definitions	Establish what customer operations covers, what an agent is permitted to be, and what resolution means	Discovery agents map current processes, channels and volumes
Assumptions	Record the assumptions: volume stability, data quality, staff acceptance, regulatory position	Monitoring agents track the indicators that would falsify each
Principles	Agree non-negotiables — escalation rights, data boundaries, accountability	Guardrails encode the principles into the execution path
Hypotheses	State the candidate designs and their predicted effects	Multi-perspective agents stress-test each from stakeholder positions
Evidence	Design the evaluation on real cases, including the difficult tail	Evaluation harnesses run continuously against held-out cases

Euclidean step	FDE action	Agentic support
Proof	Set the threshold that would justify scaling, before seeing results	Trade-off exploration maps the achievable surface
Traceability	Require a decision artefact for every consequential choice	Memory agents capture rationale, trade-offs and outcomes
Learning	Define what triggers revision and who owns it	Continuous evaluation detects drift and recommends change

The five lenses are applied across the whole of that sequence rather than at any single step. Run as a diagnostic on the same programme, they read as follows.

Lens	Applied to the customer operations programme
Structure	Which dimensions of the case mix actually drive cost and dissatisfaction — and which are we ignoring because they are hard to measure?
Uncertainty	How is volume, complexity and failure distributed? The tail, not the mean, determines whether the deployment holds
Evidence	Which results should move us: sample size, selection effects, and whether the evaluation set resembles live traffic
Relationships	How are the channels, systems, teams and escalation paths connected, and where does a local change propagate?
Information	Which single signal, if we had it, would change the decision about scope or scale? Build that first

6.3 Failure modes

Six failure modes recur with enough regularity to be treated as predictions rather than risks.

Failure mode	Description	Euclidean diagnosis
Solving the stated problem	The programme delivers exactly what was requested and does not address what was wrong	No definitions; the object was never specified
Representation failure	Rigorous work conducted on a framing that made the decisive variables invisible	The representation was inherited rather than chosen
Silent objective substitution	The metric becomes the goal and diverges from the outcome it proxied	An assumption was made and never registered
Demonstration as evidence	Confidence derived from curated demonstrations rather than measurement on real cases	A diagram was mistaken for a proof
Governance by document	Controls exist in policy and not in the execution path	The system is internally inconsistent

Failure mode	Description	Euclidean diagnosis
Delegated judgement	Decision and adaptation are quietly absorbed by the agent because no human stage was designed	The soundness of the producing system was never examined

6.4 Recommendations

1. Begin every consequential initiative with definitions. Require that the key terms — success, risk, quality, autonomy — carry agreed measurements before design begins. The cost of this step is days; the cost of skipping it is quarters.
2. Maintain an assumption register with failure indicators. For each load-bearing assumption, record what observation would show it to be false and who is watching for it.
3. State organisational axioms explicitly. Where strategic disagreement persists at the tactical level, escalate it to the level of principle, which is where it actually resides.
4. Choose the representation deliberately, and record the alternatives rejected. Before committing to a framing of the problem, state at least two others and say what each would make visible. Representation is a decision, and an inherited representation is a decision taken by someone else.
5. Run the five lenses as a standing diagnostic. Structure, uncertainty, evidence, relationships, information — applied to any consequential initiative, they expose omissions far faster than a review board does. Their principal value is identifying the question nobody asked.
6. Set the evidence threshold before running the experiment. Decide in advance what result would justify scaling, and hold to it. Retrospective thresholds are not evidence.
7. Build governance into the execution path. Any control that cannot be pointed to in the running system should be treated as absent.
8. Design the human stages deliberately. Specify where decision and adaptation occur, who owns them, and what triggers escalation — before the workflow is built, not after an incident.
9. Invest in organisational memory ahead of additional automation. The compounding return comes from the second loop — projects to lessons to patterns to standards — not from the tenth agent.
10. Develop the Forward Deployed Engineer capability as an institutional asset. The role is the mechanism by which method reaches operations; treating it as contract delivery capacity forfeits the compounding entirely.
11. Treat mathematical foundations as a leadership capability rather than a technical prerequisite. The derivations are delegable; the lenses are not. What cannot be outsourced is the ability to recognise, in an unfamiliar situation, which dimensions matter and which information would change the decision.

6.5 Conclusion

Execution is becoming abundant. Judgement is not. As the cost of doing falls towards zero, the binding constraint on organisational performance shifts to deciding — what to do, why, under

what constraints, and how one would know it was working. This is not a temporary condition of the current technology cycle; it is the structural consequence of automating a category of work.

The response is not to resist the automation, nor to await a model capable of supplying the judgement. It is to become deliberate about the part that remains. Euclid demonstrated, twenty-three centuries ago, that reasoning can be made inspectable: that a body of knowledge can be built so that every conclusion carries its lineage back to something explicitly assumed. The mathematics of artificial intelligence supplies the lenses through which the content of that reasoning becomes visible — structure, uncertainty, evidence, relationships and information. The Forward Deployed Engineer is the role that brings both into contact with organisational reality. Agentic workflows are the infrastructure that lets the whole operate continuously rather than episodically.

Define carefully. Assume explicitly. Reason systematically. Validate rigorously. Build incrementally. The tools have changed beyond recognition; the method has not needed to.

There is a final observation worth stating plainly. The mathematics that made this generation of AI possible has turned out to be more valuable as a way of thinking than as a body of technique. The technique is now largely delegable. The way of thinking is not — and the organisations that grasp this will treat their mathematical foundations not as the entry requirement for building models, but as the working method by which their most difficult decisions are taken.

The organisations that prosper in the next decade will not be distinguished by the quality of their models. Models will be commodity infrastructure, procured rather than built. They will be distinguished by the quality of the higher-order reasoning they can perform reliably, repeatedly, and at institutional scale — and by whether that reasoning survives the departure of the people who first performed it.

About this paper

This position paper forms part of a body of work at Erdos Research - which is ongoing. Specifically, we intend to extend these ideas to AI Skills, AI for Deeptech and more details into the use of Agentic workflows to redefine processes. The work is primarily created by Ajit Jaokar with contributions as indicated. It is released under the MIT license. The primary goal is educational - and is implemented in our work and teaching.

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